

Information Communication Technology

Course: CS — Certified Secretaries

Level: Foundation Level (CS16)

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Syllabus version: Based on the standard KASNEB CS Foundation Level Information Communication Technology syllabus topics (CS16) — computer fundamentals, common office/governance software, data security, and electronic communication, oriented toward the needs of a company secretary. Authored from general ICT knowledge and standard syllabus topic coverage, not copied from any existing notes provider or study guide. Cross-check terminology and emphasis against the current KASNEB syllabus and examiner's reports before the exam.

Original work notice. Every explanation, worked example, and question in this document was written from scratch for this paper. No text has been copied or paraphrased from any KASNEB past paper, textbook, or third-party notes/study guide. The scenarios and practice questions are original.

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1. Computer hardware and software, basics

Hardware is the physical, tangible components of a computer system (input devices, output devices, the CPU, storage devices). **Software** is the programs and instructions that run on hardware, divided into **system software** (e.g. the operating system, managing hardware and providing a platform for other software) and **application software** (programs performing specific user tasks, e.g. word processing, spreadsheets).

2. Common office and governance-related software

Word processing software drafts board papers, minutes, and correspondence. Spreadsheet software supports simple financial summaries and basic data analysis. Presentation software supports board and shareholder presentations. Increasingly, dedicated **board management software** and **document management systems** support governance-specific tasks — securely distributing board papers, tracking document versions, and managing approval workflows.

3. Databases, basics

A **database** is an organised collection of structured data, allowing efficient storage, retrieval, and management of information — e.g. a company's register of members, tracking shareholder details and shareholdings, is fundamentally a structured database that must be kept accurate and readily retrievable, often as a statutory requirement.

4. Computer networks and the internet

A **computer network** connects multiple devices to share resources and information — ranging from a small **local area network (LAN)** within one office to the internet, a global network of networks. Reliable network access underpins most modern governance workflows: circulating board papers electronically, hosting virtual board meetings, and accessing cloud-based document repositories.

5. Email and electronic communication in governance

Email remains a core tool for formal business communication — circulating notices of meetings, board papers, and minutes — valued for providing a written, time-stamped, retrievable record. Good practice includes: using clear, professional subject lines and content,

confirming recipients before sending sensitive governance documents, and retaining email records in line with the organisation's records retention policy.

6. Information and data security

Information security protects data from unauthorised access, alteration, or destruction. Key measures: **access controls** (passwords, permission levels restricting who can view/edit sensitive documents), **encryption** (rendering data unreadable without proper authorisation), regular **backups** (protecting against data loss), and staff awareness of risks such as phishing (fraudulent attempts to obtain sensitive information). For a company secretary, protecting board papers and shareholder records is a governance responsibility, not merely a technical IT concern.

7. Board portals and e-governance tools

A **board portal** is a secure digital platform for distributing board papers, managing meeting schedules, and supporting board communication — replacing (or supplementing) physical board packs with a controlled, auditable digital equivalent. Benefits include faster distribution, version control (ensuring directors always view the current document), and a clear access log (useful evidence that required papers were actually circulated to all directors in good time).

8. Data protection considerations

Organisations handling personal data (of employees, shareholders, customers) are generally subject to **data protection** obligations — requiring data to be collected and used only for legitimate purposes, held securely, and not retained longer than necessary. A company secretary should be aware of these obligations at a governance level, since data protection compliance failures can expose an organisation to real regulatory and reputational risk, similar to other compliance failures the secretarial function is expected to help prevent.

9. Full worked question — evaluating an information-security weakness

Scenario. A company circulates confidential board papers to all directors via a shared, unencrypted email attachment, with no password protection and no access log of who has opened the documents.

Required: (a) Identify the information-security weakness. (b) Explain the risk this creates. (c) Recommend an improvement.

Solution:

(a) The weakness is **inadequate protection of confidential information** — sensitive board papers are being sent via an unencrypted, unprotected channel with no way to verify or control who actually accesses them.

(b) This creates a risk that confidential, market-sensitive, or otherwise commercially/legally sensitive board information could be intercepted, forwarded, or accessed by unauthorised parties, with no audit trail to establish who accessed the documents or when — a genuine governance and confidentiality failure, not merely a technical inconvenience.

(c) **Recommendation:** adopt a secure board portal (or at minimum, password-protected/encrypted documents with restricted, individually tracked access) for circulating confidential board papers, providing both stronger protection and a clear access log as evidence of proper, controlled distribution.

10. Practice examination — KASNEB-style paper (original, not a past paper)

Note on format. This section is laid out the way a KASNEB CS examination paper is laid out — a timed paper with five compulsory 20-mark questions — purely so you can practise under realistic exam conditions. Every question below is an original scenario written for this document. **It is not a reproduction of any actual KASNEB past examination paper.**

CS FOUNDATION LEVEL — INFORMATION COMMUNICATION TECHNOLOGY (PRACTICE PAPER)

TIME ALLOWED: 3 HOURS

Instructions to candidates: Answer ALL FIVE questions. Each question carries 20 marks. Show all workings clearly.

QUESTION ONE

(a) Distinguish between computer hardware and software, with TWO examples of each. (12 marks)

(b) Distinguish between system software and application software. (8 marks)

(Total: 20 marks)

QUESTION TWO

(a) Define "database" and explain its relevance to a company's register of members. *(10 marks)*

(b) Explain the role of computer networks in supporting modern governance workflows. *(10 marks)*

(Total: 20 marks)

QUESTION THREE

(a) Outline FOUR information-security measures relevant to protecting confidential board papers. *(16 marks)*

(b) Explain what "phishing" means. *(4 marks)*

(Total: 20 marks)

QUESTION FOUR

(a) Define "board portal" and outline THREE benefits it offers over physical board packs. *(12 marks)*

(b) Explain why data protection compliance is relevant to a company secretary at a governance level. *(8 marks)*

(Total: 20 marks)

QUESTION FIVE

A company circulates confidential board papers via an unencrypted, unprotected shared email attachment, with no access log.

(a) Identify the weakness in this approach. *(6 marks)*

(b) Explain the risk this creates. *(8 marks)*

(c) Recommend an improved approach. *(6 marks)*

(Total: 20 marks)

11. Marking guide — model answers to the practice examination**QUESTION ONE**

(a) *(12 marks)* — Hardware: physical components, e.g. keyboard, monitor. Software: programs/instructions, e.g. word processor, operating system.

(b) (8 marks) — System software manages hardware and provides a platform for other software (e.g. an operating system); application software performs specific user tasks (e.g. word processing).

QUESTION TWO

(a) (10 marks) — A database is an organised collection of structured data enabling efficient storage/retrieval/management; a register of members is fundamentally a structured database of shareholder details that must be kept accurate and readily retrievable, often as a statutory requirement.

(b) (10 marks) — Networks (including the internet) enable electronic circulation of board papers, virtual board meetings, and access to cloud-based document repositories, underpinning most modern governance workflows.

QUESTION THREE

(a) (16 marks — 4 marks each) — Access controls/passwords, encryption, regular backups, staff awareness of risks like phishing.

(b) (4 marks) — Phishing is a fraudulent attempt to obtain sensitive information (e.g. passwords) by disguising as a trustworthy source, typically via email.

QUESTION FOUR

(a) (12 marks) — A board portal is a secure digital platform for distributing board papers and supporting board communication; benefits: faster distribution, version control, a clear access log (any three, 4 marks each, plus definition).

(b) (8 marks) — Data protection compliance failures expose the organisation to regulatory and reputational risk; a company secretary should be aware of these obligations at a governance level, similar to other compliance failures the secretarial function helps prevent.

QUESTION FIVE

(a) (6 marks) — Inadequate protection of confidential information — sent via an unencrypted, unprotected channel with no access verification.

(b) (8 marks) — Risk of interception or unauthorised access to sensitive board information, with no audit trail to establish who accessed it or when — a genuine confidentiality/governance failure.

(c) (6 marks) — Adopt a secure board portal, or at minimum password-protected/encrypted documents with restricted, individually tracked access.

12. Exam technique and revision checklist

- Hardware vs software, system vs application software: know these distinctions precisely with concrete examples — frequently and specifically tested.
- Security-measure questions: name specific measures (access controls, encryption, backups, staff awareness) rather than a vague "keep it secure."
- Board portal / e-governance questions: always connect the technology back to a specific governance benefit (version control, access log, faster distribution), since that's the CS-specific angle these questions test.
- Scenario questions: identify the weakness, explain the risk, then give a specific, practical recommendation — the standard three-part structure across CS scenario questions.
- Data protection questions: connect compliance failures to regulatory/reputational risk, framed as a governance concern, not just a technical IT issue.

Quick revision — one line per topic:

1. Hardware (physical) vs software (programs); system software (manages hardware) vs application software (user tasks).
2. Database: organised, structured data — e.g. a register of members.
3. Networks/internet: enable electronic board paper circulation, virtual meetings, cloud repositories.
4. Email: formal communication, provides a written, time-stamped record.
5. Information security: access controls, encryption, backups, staff awareness of phishing.
6. Board portal: secure digital platform — faster distribution, version control, access log.
7. Data protection: legitimate use, secure holding, not retained longer than necessary.
8. CS's ICT role: governance-level oversight of confidentiality and compliance, not deep technical IT work.

End of study notes. This document is intended as original exam-preparation material and should be used alongside the current official KASNEB syllabus and examiner's reports for the paper.